Technical Document #829: Cybersecurity - Comprehensive Overview

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Multi-factor authentication (MFA) requires multiple forms of verification. It significantly reduces the risk of unauthorized access.

Intrusion detection systems (IDS) monitor network traffic for suspicious activity. They alert administrators of potential security breaches.

Security information and event management (SIEM) systems collect and analyze security logs. They provide real-time visibility into security events.

Social engineering manipulates people into revealing confidential information. Phishing is the most common social engineering attack.

Zero trust architecture assumes no implicit trust and verifies every request. It minimizes the attack surface and limits lateral movement.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Penetration testing simulates cyber attacks to identify vulnerabilities.
- Encryption converts plaintext into ciphertext to protect data confidentiality.
- Incident response plans define how organizations respond to security incidents.